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Osteo Sentinel Embedded and IoT system for real-time Osteoporosis fracture risk prediction using Machine learning

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ABSTRACT

Osteoporosis is a bone disease that increases fracture risk, especially in elderly individuals. This project presents an embedded and IoT-based system for real-time osteoporosis fracture risk prediction using Machine Learning. Multiple sensors, including accelerometer, pulse, and SpO₂ sensors, collect physiological and motion data through an ESP32 microcontroller. The data is transmitted via the Internet of Things to a server where a trained model predicts risk levels as low, medium, or high. The system also provides food and doctor recommendations, along with real-time alerts, enabling early detection, continuous monitoring, and improved preventive healthcare management.

INTRODUCTION

- Osteoporosis is a chronic bone disease that reduces bone density and increases the risk of fractures, especially among elderly individuals and postmenopausal women. These fractures can lead to severe health complications, disability, and reduced quality of life.
- Traditional diagnostic methods, such as bone density scans, are typically performed in hospitals and do not support continuous monitoring or early real-time risk detection. With advancements in Internet of Things and Machine Learning, smart healthcare systems can now provide continuous monitoring and predictive analysis.

PROBLEM STATEMENT

- No real-time fracture risk monitoring system is available. Patients are unable to track their bone health continuously.
- Delayed diagnosis increases chances of severe fractures. Lack of early warning leads to critical conditions.
- Existing systems do not provide personalized recommendations. Users do not receive preventive guidance.
- Need for a cost-effective and portable solution. System should be accessible for daily usage.

LITERATURE SURVEY

S.NO	AUTHOR NAMES	PUBLISHED NAME	LIMITATIONS
1	Rajpurkar et al. (2017), proposed a deep learning model (CheXNet) for detecting abnormalities in X-ray images. The model uses CNN to automatically extract features and improve diagnostic accuracy.	IEEE Transactions on Medical Imaging	Requires large annotated datasets and high computational resources.
2	Kim & MacKinnon (2018), developed a CNN-based system for bone fracture detection from X-ray images, achieving high accuracy in classification tasks.	Elsevier – Expert Systems with Applications	Performance decreases with low-quality or noisy images.
3	Hassan et al. (2020), applied transfer learning models like VGG 16 and ResNet for fracture detection, improving classification performance with pre-trained networks.	Springer – Journal of Medical Systems	Requires fine-tuning and may suffer from overfitting on small datasets.
4	Tanzi et al. (2019), used Region-based CNN (R-CNN) for detecting and localizing fractures in medical images. The system improves detection precision.	IEEE Access	High computational cost and longer training time.
5	Patil & Sharma (2023), proposed an AI + IoT-based bone fracture detection system integrating CNN with NodeMCU, OLED, and alert system for real-time monitoring.	IEEE – International Conference on AI in Healthcare	IoT integration may introduce latency and depends on network reliability.

EXISTING SYSTEM

- Uses ensemble machine learning (Super Learner) for prediction. It combines multiple models to improve accuracy and reliability.
- Integrates genomic and clinical data for analysis.
- Lack of interpretability in ensemble models. It is difficult for doctors to understand how predictions are made.
- Performance may decrease over time without proper maintenance. Privacy and security concerns with genomic data.

LIMITATIONS OF EXISTING SYSTEM

- Expensive and not affordable for common users. It limits the widespread usage of the system.
- No real-time monitoring capability. Continuous health tracking is not possible.
- Lack of an early warning system and the Fracture risks are detected too late.
- No personalised healthcare suggestions. Patients do not receive preventive advice.
- Bulky design reduces portability. It is not suitable for wearable applications.

PROPOSED SYSTEM

- Embedded system using ESP32 for real-time processing. It acts as the central controller of the system.
- Multiple sensors collect physiological and motion data. This improves accuracy of prediction.
- Data is transmitted to cloud using IoT technology. Enables remote monitoring and storage.
- Machine learning model predicts fracture risk levels. Provides fast and reliable classification.
- Smart alerts and recommendations are generated. Helps in early intervention and prevention.

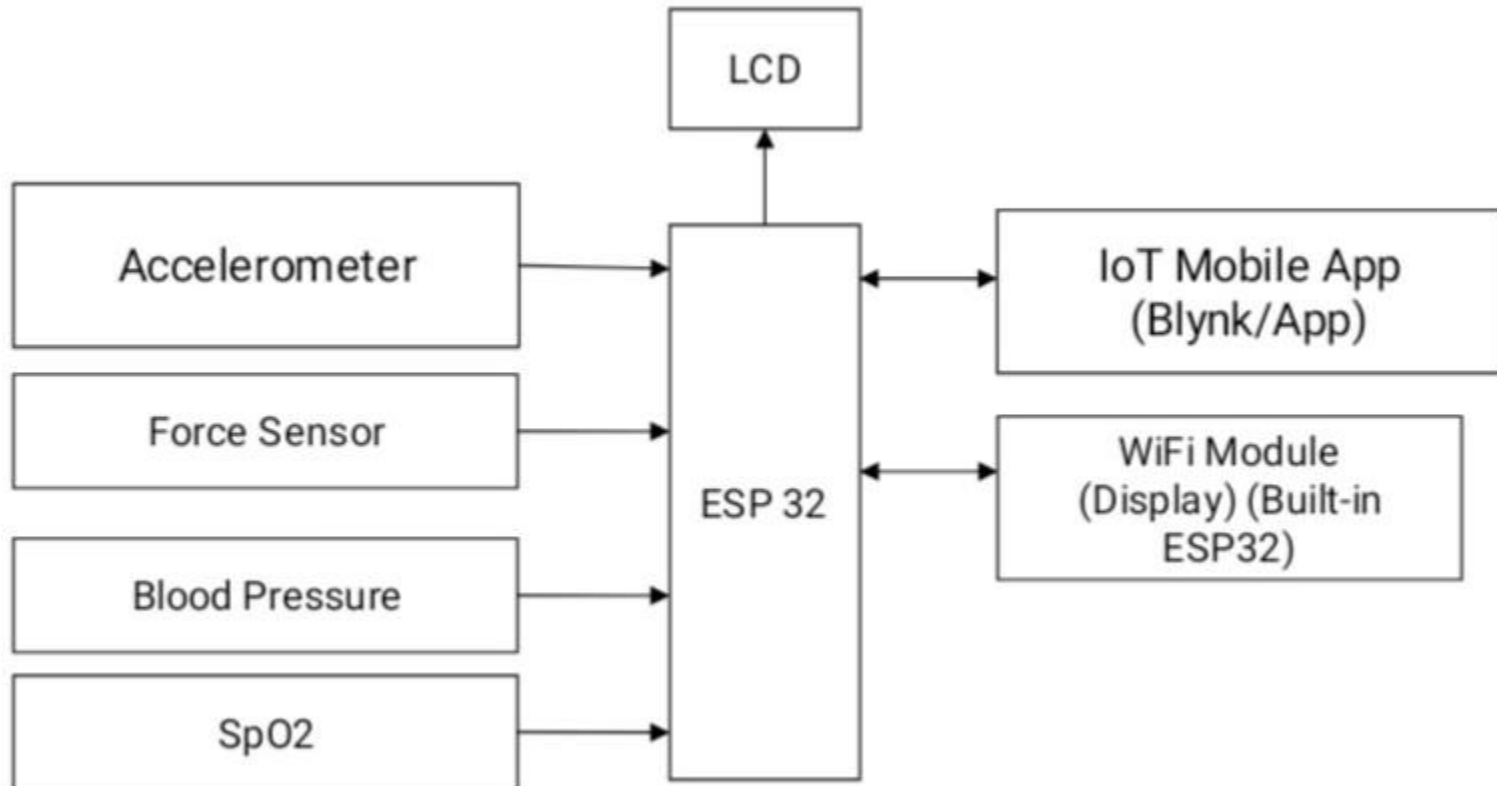
HARDWARE COMPONENTS

- ESP32
- Accelerometer(MPU 6050)
- Force sensor(FSR)
- Blood Pressure sensor
- SpO2 sensor(MAX 30100/MAX30102)
- LCD display(16*2)

SOFTWARE COMPONENTS

- **Programming Language:**
Embedded C
Python
- **Integrated Development Environment (IDE):**
Arduino IDE
Visual Studio Code
- **IoT Platform:**
Blynk
- **Database / Cloud Platform:**
Firebase
AWS Cloud
- **Machine Learning Tools:**
Scikit-learn
NumPy / Pandas

BLOCK DIAGRAM



WORKING PRINCIPLE

- Sensors continuously collect physiological and motion-related data from the user
- ESP32 processes the collected data in real time.
- Data is transmitted to the cloud using Wi-Fi (IoT connectivity)
- Preprocessing and feature extraction are applied to improve data quality
- Machine learning model (Random Forest) analyzes data and predicts fracture risk
- Output is displayed on LCD and mobile app with alerts for high-risk conditions.

TECHNOLOGY

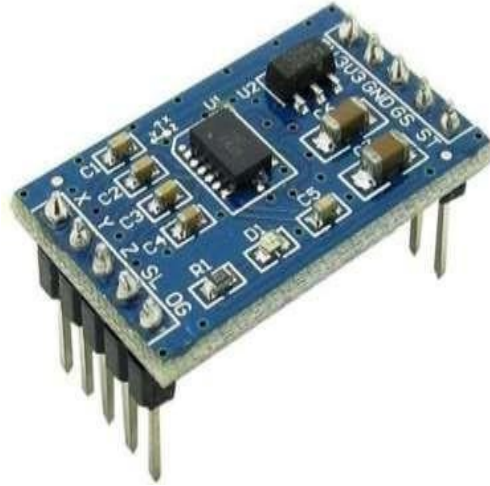
- Embedded systems are used for real-time data acquisition from sensors.
- Internet of Things enables wireless communication and remote monitoring.
- Machine Learning (Random Forest) is used for accurate fracture risk.
- Cloud computing stores and processes large volumes of sensor data.
- Mobile application technology provides a real-time user interface and alerts.
- Wireless communication (Wi-Fi via ESP32) ensures data transmission.

ESP32 Microcontroller



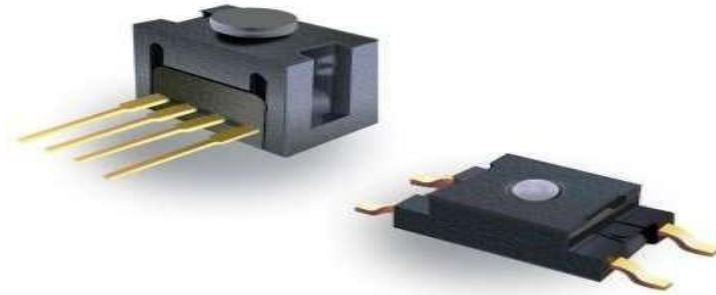
- Microcontroller with built-in Wi-Fi capability for IoT connectivity
- Operating voltage: 3.3V
- Digital I/O pins: Up to 34 GPIO
- Communication protocols: UART, SPI, I2C, PWM
- Application: Acts as the central controller for collecting sensor data, performing communication, and sending data to the cloud for Machine Learning analysis

Accelerometer Sensor(MPU 6050)



- Type: 3-axis accelerometer and gyroscope
- Operating voltage: 3.3V–5V
- Communication: I2C protocol
- Measurement: Acceleration and motion detection
- Application: Detects falls and sudden movements, which are key indicators in Osteoporosis fracture risk

Force sensor(FSR)



- Type: Pressure sensing resistor
- Operating voltage: 3.3V–5V
- Output: Analog
- Sensitivity: Varies with applied force
- Application: Measures pressure/load changes to analyze body stability and impact

Blood Pressure Sensor



- Type: Digital BP monitoring module
- Operating voltage: 5V
- Measurement: Systolic and diastolic pressure
- Output: Digital
- Application: Monitors blood pressure variations, which are important for health risk analysis

SpO₂ Sensor(MAX 30100)



- Type: Pulse oximeter sensor
- Operating voltage: 3.3V
- Communication: I2C
- Measurement: Oxygen saturation (SpO₂) and pulse rate
- Application: Monitors blood oxygen levels to assess overall health condition

LCD Display



- Display type: Character LCD
- Operating voltage: 5V
- Interface: I2C / Parallel
- Character format: 16×2
- Viewing angle: Wide
- Application: Displays real-time sensor readings and fracture risk levels (Low / Medium / High)

SOFTWARE DESCRIPTION

- Sensor data is collected and transmitted using embedded programming
- Data preprocessing removes noise and normalizes input values
- Feature extraction is performed for machine learning analysis
- Random Forest algorithm is used for fracture risk prediction
- Cloud platform stores and processes real-time data
- Mobile application displays results to the user

INTERFACE DESIGN

- LCD screen displays real-time health parameters clearly
- Mobile application provides detailed values in dashboard
- Risk levels shown as Low, Medium and High categories
- Graphical visualization helps track health trends over time
- User-friendly design ensures easy navigation and accessibility
- Alert notifications are triggered during high-risk conditions.

ALGORITHM USED

RANDOM FOREST:

- Random Forest is a machine learning algorithm used for classification and prediction. It works by creating multiple decision trees and combining their results to improve accuracy and reduce errors.
- Random Forest consists of the following components:
 - Decision Trees
 - Bootstrap Sampling
 - Feature Randomisation
 - Majority Voting
- Each decision tree analyses input data and predicts the risk level. The final output is determined by majority voting among all trees.
- As the number of trees increases, prediction becomes more accurate and stable. It effectively classifies fracture risk into Low, Medium, and High level.

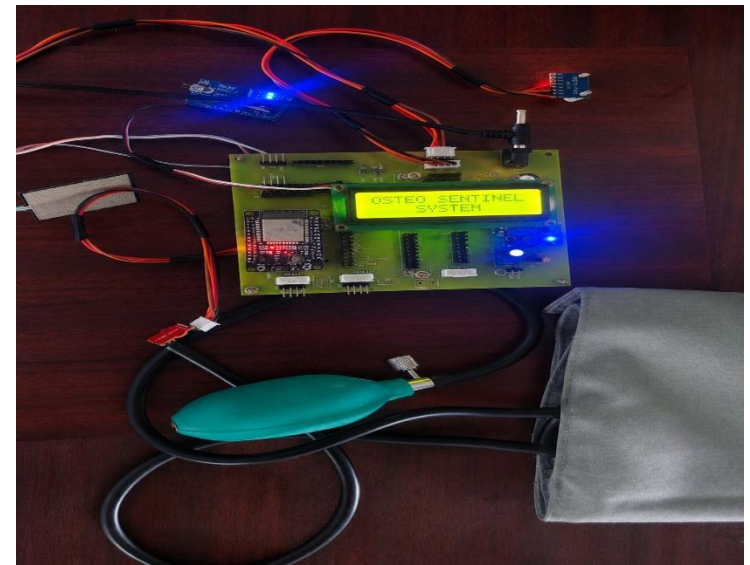
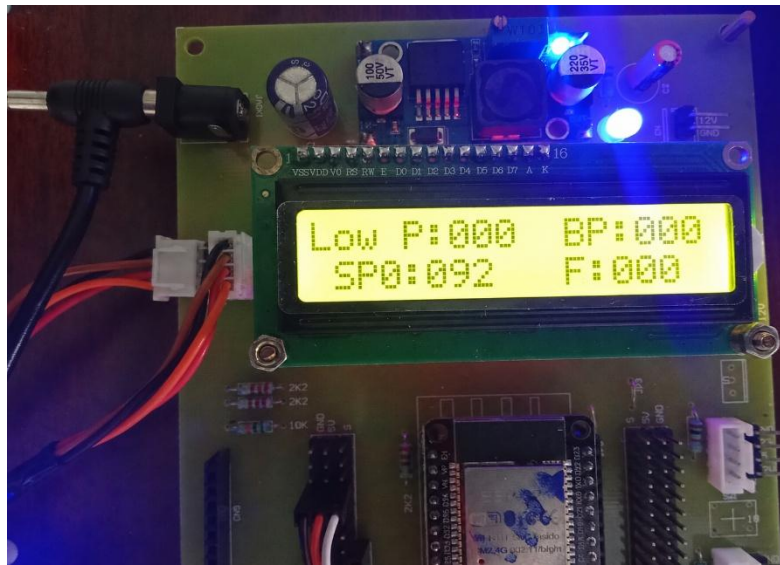
ADVANTAGES

- Provides real-time health monitoring. Enables early detection of fracture risk.
- Cost-effective and portable system design. Suitable for everyday use.
- Continuous data tracking improves accuracy. Helps in better decision making.
- Personalized recommendations enhance healthcare. Users receive preventive guidance.
- Easy to use and accessible system. Improves user experience.

APPLICATIONS

- Used for real-time fracture risk monitoring in elderly patients. Helps in early detection and prevention of bone-related injuries.
- Applicable in home healthcare systems.
- Useful in rehabilitation and post-fracture recovery.
- Beneficial in smart healthcare and telemedicine systems. Enhances digital healthcare infrastructure and services.

OUTPUT SCREENSHOTS



Osteoporosis Fracture Risk Prediction

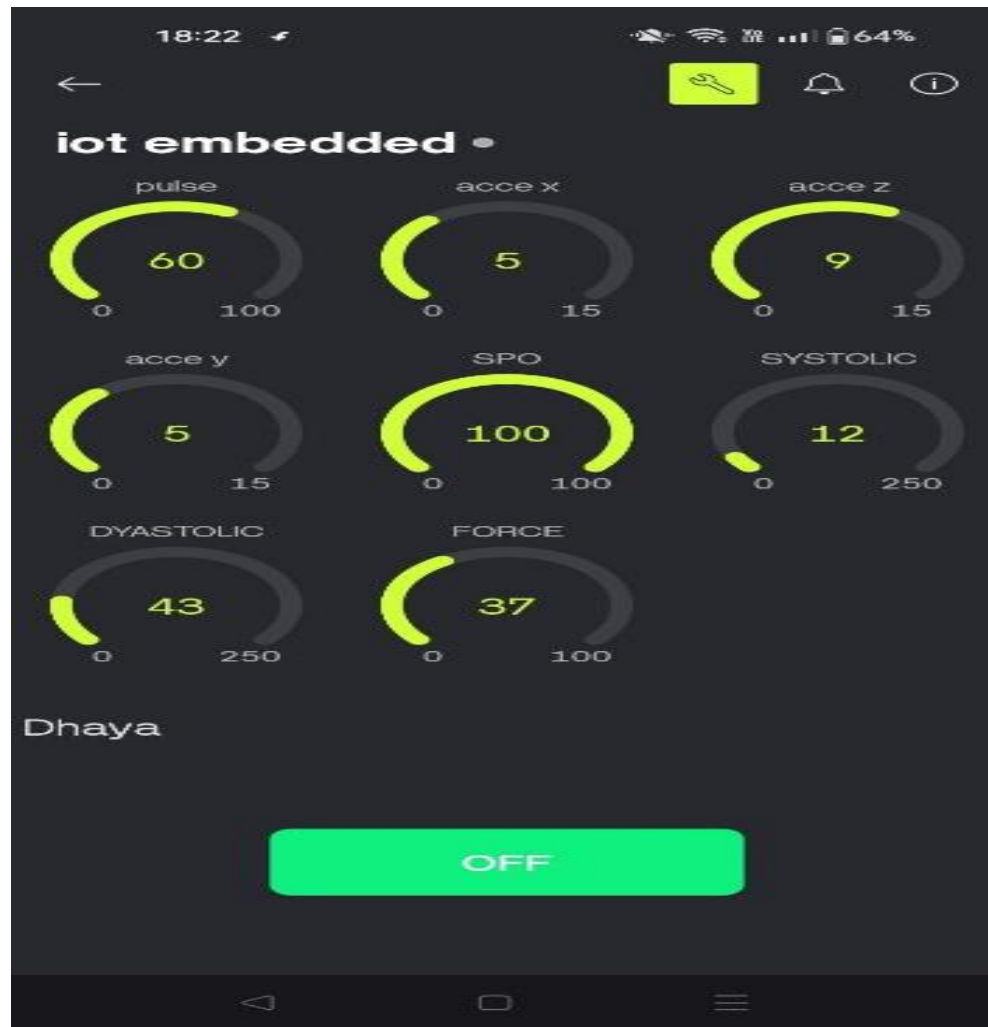
Fetch Real-Time Data & Predict

History Data

	pulse	accel_x	accel_y	accel_z	bp_diastolic	bp_systolic	spo2	force	result	food	doctor	time
0	0	9	5	7	0	0	0	0	MEDIUM	Cheese, Spinach, Broccoli, Salmon	Orthopedic Doctor	2026-03-31 13:59:41.698104
1	0	9	5	7	0	0	0	0	MEDIUM	Cheese, Spinach, Broccoli, Salmon	Orthopedic Doctor	2026-03-31 14:04:41.673867
2	60	5	5	9	43	12	100	37	MEDIUM	Cheese, Spinach, Broccoli, Salmon	Orthopedic Doctor	2026-05-08 13:17:25.394791
3	60	5	5	9	43	12	100	37	MEDIUM	Cheese, Spinach, Broccoli, Salmon	Orthopedic Doctor	2026-05-08 13:23:55.841808
4	60	5	5	9	43	12	100	37	MEDIUM	Cheese, Spinach, Broccoli, Salmon	Orthopedic Doctor	2026-05-08 13:32:24.815675
5	60	5	5	9	43	12	100	37	MEDIUM	Cheese, Spinach, Broccoli, Salmon	Orthopedic Doctor	2026-05-08 13:51:45.438278
6	60	5	5	9	43	12	100	37	MEDIUM	Cheese, Spinach, Broccoli, Salmon	Orthopedic Doctor	2026-05-08 13:56:53.682106

Download CSV

Osteoporosis Fracture Risk Prediction Dashboard



Real-Time Health Monitoring Dashboard in Blynk Application

OSTEOPOROSIS DATASET

S.No	pulse	accel_x	accel_y	accel_z	bp_diastolic	bp_systolic	spo2	force	risk
1	89	1.0760411065600568	4.518532692765021	4.794589977147839	73	107	100	18	LOW
2	109	6.20032141798068	7.75719171268763	5.53085951208228	84	135	95	61	MEDIUM
3	114	14.95266416547274	11.322291350885362	14.36019595594211	99	152	81	89	HIGH
4	62	1.083804801402671	3.384895050131779	0.3039230081683591	71	112	96	12	LOW
5	104	9.43153903435318	9.7724899214395	5.976312225682942	92	124	94	40	MEDIUM
6	129	11.916176982854608	14.038009824823613	13.886928882474695	109	164	84	86	HIGH

CONCLUSION

- System enables real-time fracture risk prediction. Improves preventive healthcare.
- Combines IoT, embedded systems, and machine learning. Enhances system performance.
- Provides early detection and smart recommendations. Reduces severe complications.
- Supports remote monitoring and continuous tracking. Ensures patient safety

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- World Health Organization (WHO), “Assessment of Osteoporosis at the primary health care level,” WHO Press.